



NORTHERN UNIVERSITY OF BUSINESS AND TECHNOLOGY KHULNA

COURSE CODE: CSE 4204

COURSE TITLE: Mobile Computing Lab

[Project Name: AI Expense Analyzer]

Submitted by – Team: CSE4204-8A-T04 Naim Sheikh ID: 11210320634 Ayon Banerjee ID: 11220120820 Md Shakinul Islam ID: 11220120822 Afia Rahman ID: 11220320870	Submitted to – Md. Riaz Mahmud Assistant Professor Department of Computer Science and Engineering Northern University of Business and Technology Khulna
---	--

Date of Submission – 22/06/2026

Index

1. Team Information.....	(2)
1.1 Team Information.....	(2)
2. Project Information.....	(3)
2.1 Overview... ..	(3)
2.2 Project Information.....	(3)
3. Screen Descriptions for UI/UX Documentation.....	(4-15)
4. User Flow Diagram.....	(16)
5. Development Roadmap, Team Task Distribution, Final Technology Stack.....	(17)

AI Expense Analyzer

1. Team Information:

Team Name: CSE4204-8A-T04

Section: 8A

Project Title: AI Expense Analyzer

Team Leader: Md. Shakinul Islam

GitHub Repository: <https://github.com/shakinul-islam/AI-Expense-Analyzer>

1.1 Team Information:

Serial	Name	Student ID	Technology	Responsibility
1	Md Shakinul Islam	11220120822	MongoDB	Database Management & Testing
2	Naim Sheikh	11210320634	Flutter	Frontend Development & UI Design
3	Ayon Banerjee	11220120820	Node.js + Express.js	Backend Development
4	Afia Rahman	11220320870	Gemini Ap	AI Integration

2. Project Information:

2.1 Overview:

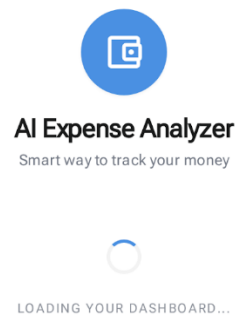
The AI Expense Analyzer is a cross-platform intelligent asset ecosystem utilizing a Flutter frontend client, managed by an asynchronous Node.js backend layered over Express.js middleware, and persistent within a MongoDB document database. To upgrade past standard, static digital ledger interfaces that require manual evaluation, the environment integrates the advanced Google Gemini API. This enables automated transaction pattern tagging, real-time localized optimization advice, predictive budgeting models, and algorithmic waste tracking to instill long-term fiscal discipline.

2.2 Project Information:

Title	Description
Project Name	AI Expense Analyzer
Problem	Users cannot understand their spending patterns from raw transaction data.
Solution	AI-powered financial insights, automated categorization, predictive budgeting.
Target Users	Individual consumers who want to save money and track expenses.
Platform	Cross-platform mobile (Android + iOS)

Screen Descriptions for UI/UX Documentation

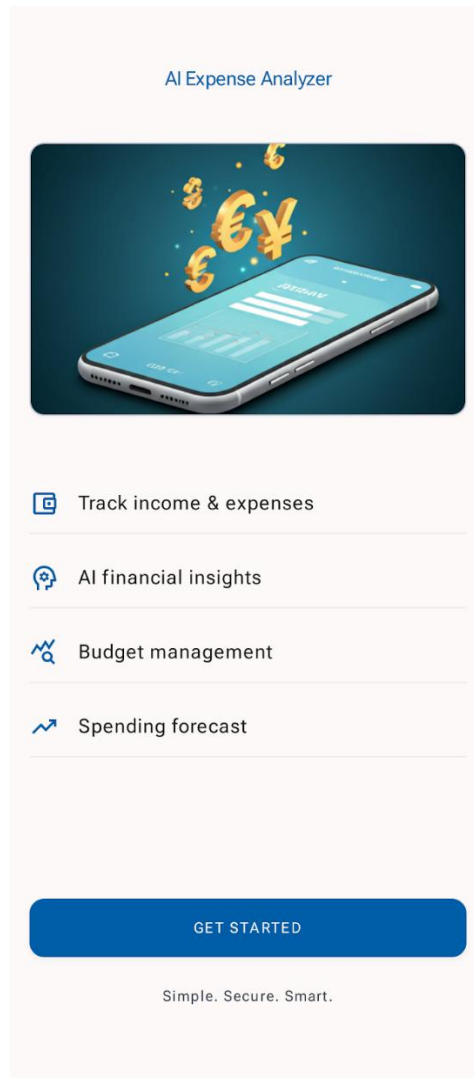
Splash Screen:



Description:

The entry point of the application. It displays the app logo and tagline, "Smarter tracking for better savings," while the system initializes and checks for existing user sessions.

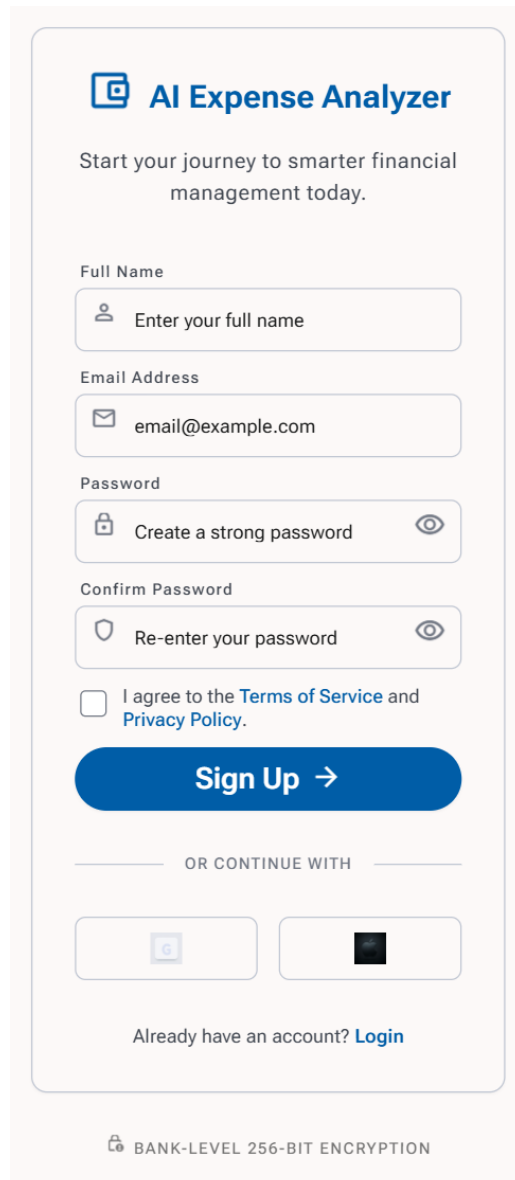
Onboarding_Screen:



Description:

An introductory walkthrough that highlights key features such as income/expense tracking, AI-driven financial insights, and budget management to guide new users.

Sign Up_Screen:



The sign-up screen for the AI Expense Analyzer features a clean, modern design. At the top, the app's logo and name are displayed, followed by a motivational message. The registration form includes fields for a full name, email address, and a password, with a confirmation field for the password. A checkbox for terms and privacy policy is located below the password fields. A prominent blue 'Sign Up' button is centered below the form. Below the button, users are given the option to continue with Google or Apple. A 'Login' link is provided for existing users. At the bottom, a security assurance is provided.

AI Expense Analyzer

Start your journey to smarter financial management today.

Full Name

Enter your full name

Email Address

email@example.com

Password

Create a strong password

Confirm Password

Re-enter your password

☐ I agree to the [Terms of Service](#) and [Privacy Policy](#).

Sign Up →

OR CONTINUE WITH

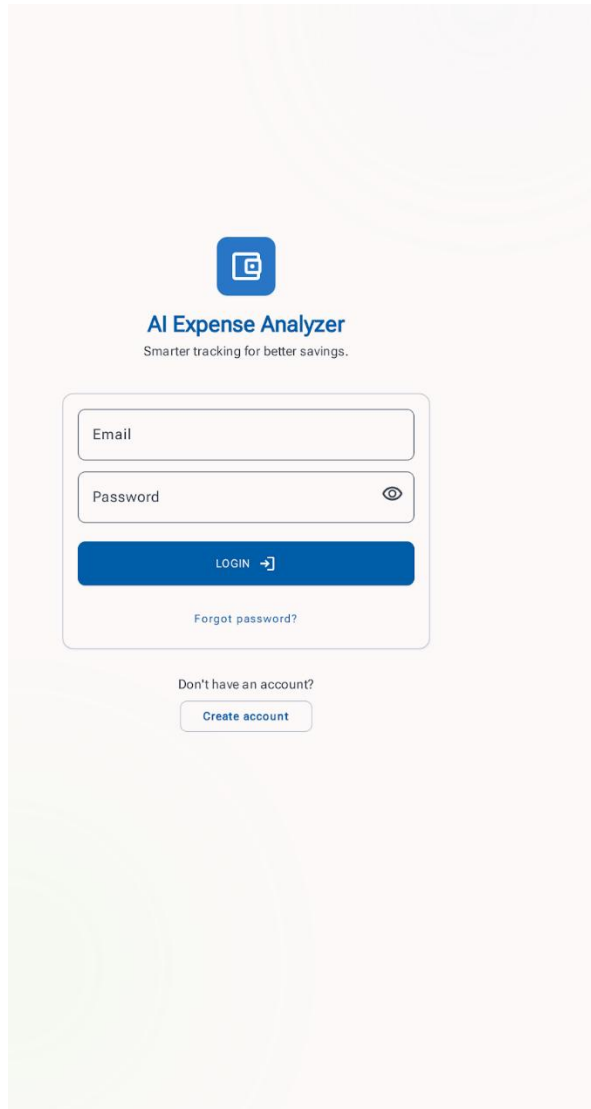
Already have an account? [Login](#)

 BANK-LEVEL 256-BIT ENCRYPTION

Description:

The registration interface where new users enter their full name, email, and password to create a secure account.

Login_Screen:



The login screen for the AI Expense Analyzer application. It features a blue square logo with a white document icon. Below the logo, the text "AI Expense Analyzer" is displayed in a bold, dark blue font, followed by the tagline "Smarter tracking for better savings." in a smaller, grey font. The login form is enclosed in a light grey rounded rectangle and contains an "Email" input field, a "Password" input field with a toggle eye icon, a blue "LOGIN" button with a right-pointing arrow, and a "Forgot password?" link. Below the login form, there is a "Don't have an account?" link and a "Create account" button.



AI Expense Analyzer
Smarter tracking for better savings.

Email

Password 

LOGIN 

[Forgot password?](#)

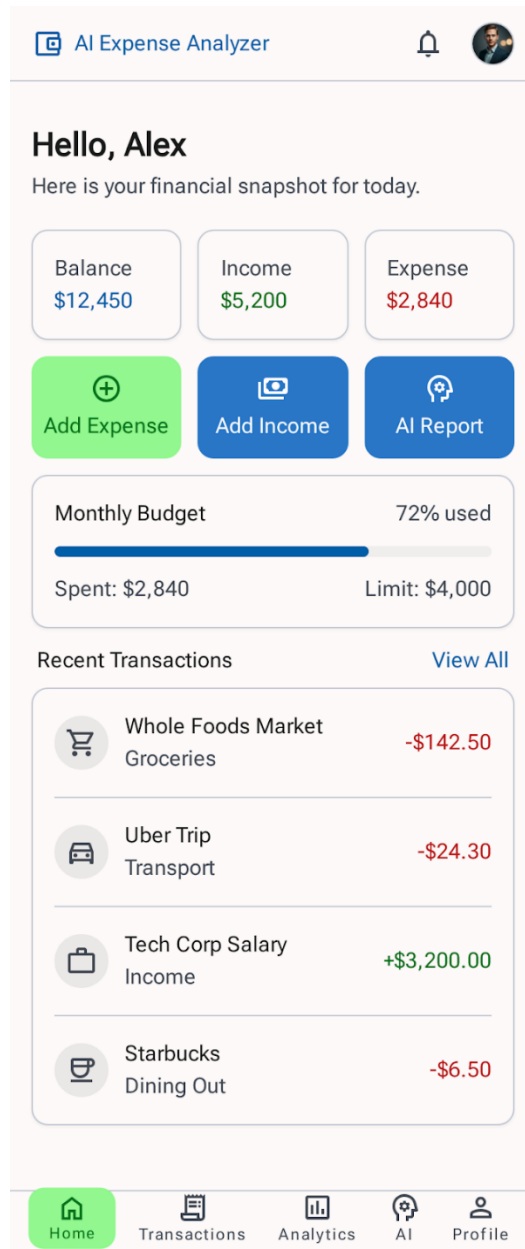
Don't have an account?

[Create account](#)

Description:

The authentication gateway where registered users provide their credentials to access their secure financial data.

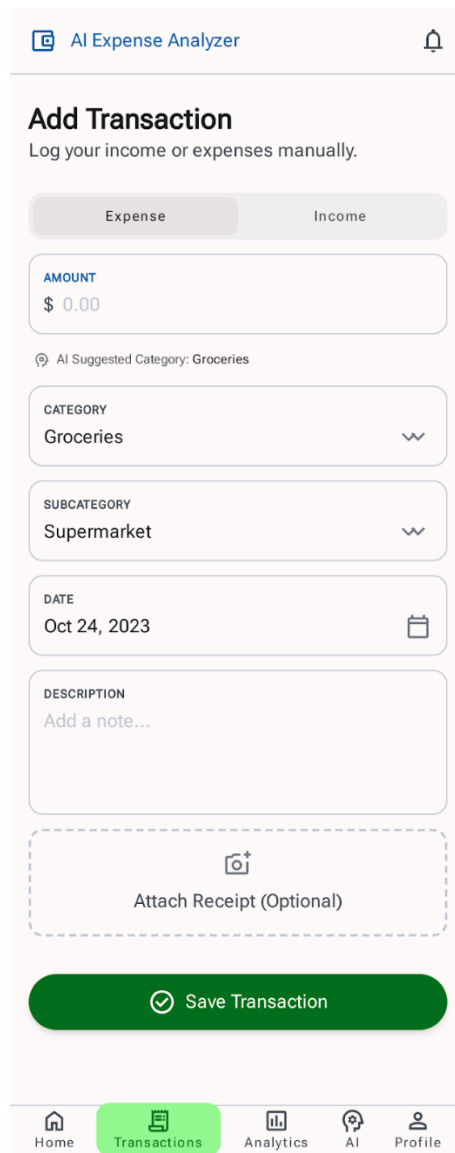
Home_Screen:



Description:

The primary dashboard providing a high-level financial snapshot, including current balance, income, expenses, and recent transaction history.

Add Transaction_Screen:



The image shows a mobile application screen titled "Add Transaction" under the header "AI Expense Analyzer". The screen is designed for logging income or expenses. It features a toggle for "Expense" (selected) and "Income". Below this is a large input field for the "AMOUNT" with a placeholder "\$ 0.00". An AI suggestion "AI Suggested Category: Groceries" is displayed. There are two dropdown menus for "CATEGORY" (set to "Groceries") and "SUBCATEGORY" (set to "Supermarket"). A "DATE" field shows "Oct 24, 2023" with a calendar icon. A "DESCRIPTION" field has a placeholder "Add a note...". An optional receipt attachment area is shown with a camera icon and the text "Attach Receipt (Optional)". A prominent green button at the bottom says "Save Transaction". The bottom navigation bar includes icons for Home, Transactions (highlighted), Analytics, AI, and Profile.

AI Expense Analyzer

Add Transaction

Log your income or expenses manually.

Expense Income

AMOUNT
\$ 0.00

AI Suggested Category: Groceries

CATEGORY
Groceries

SUBCATEGORY
Supermarket

DATE
Oct 24, 2023

DESCRIPTION
Add a note...

Attach Receipt (Optional)

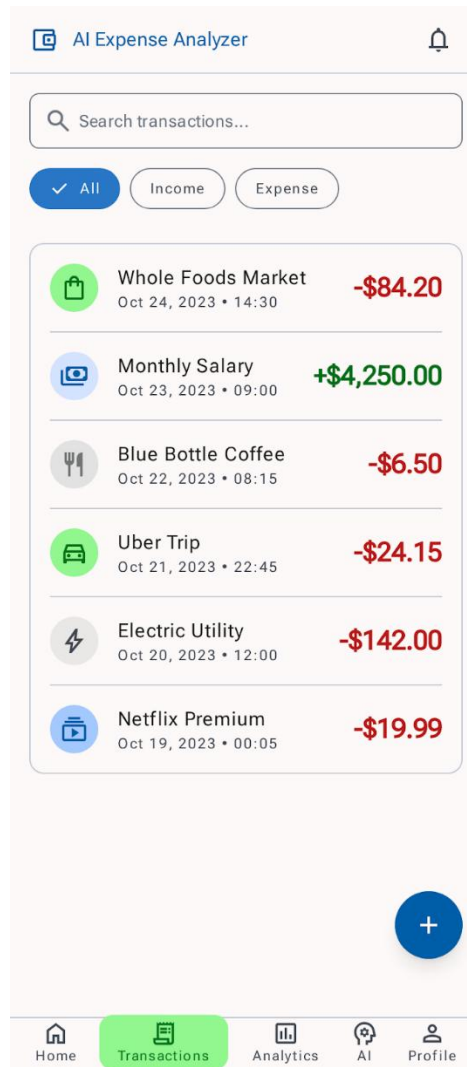
Save Transaction

Home Transactions Analytics AI Profile

Description:

A data-entry interface used for logging new income or expenses, including fields for amount, date, description, and categorization.

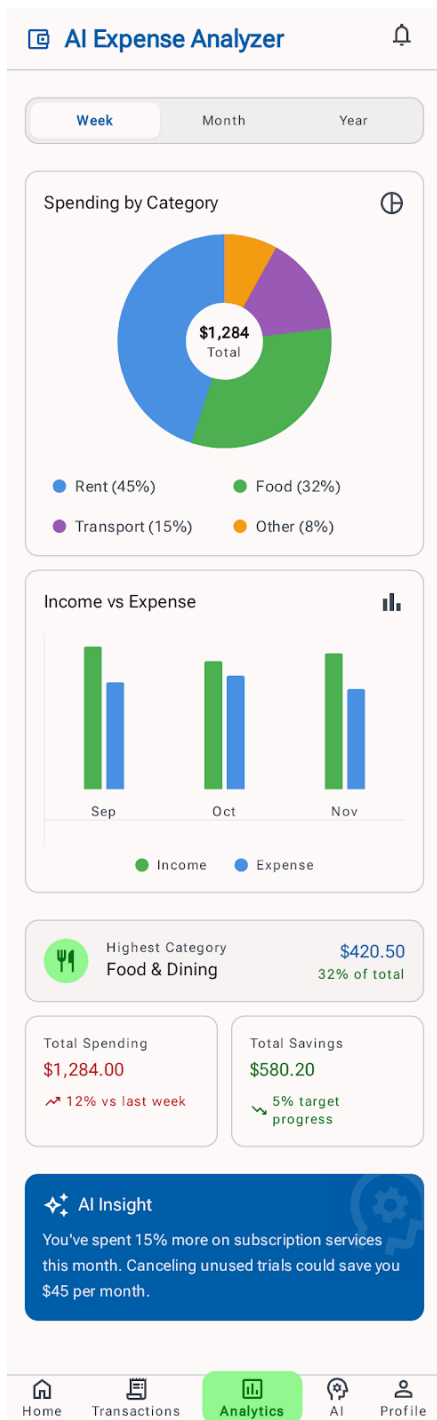
Transaction History Screen:



Description:

This is the dedicated ledger view where users can see a comprehensive, scrollable list of all past income and expense records, filterable by date, category, or amount for easier auditing and management.

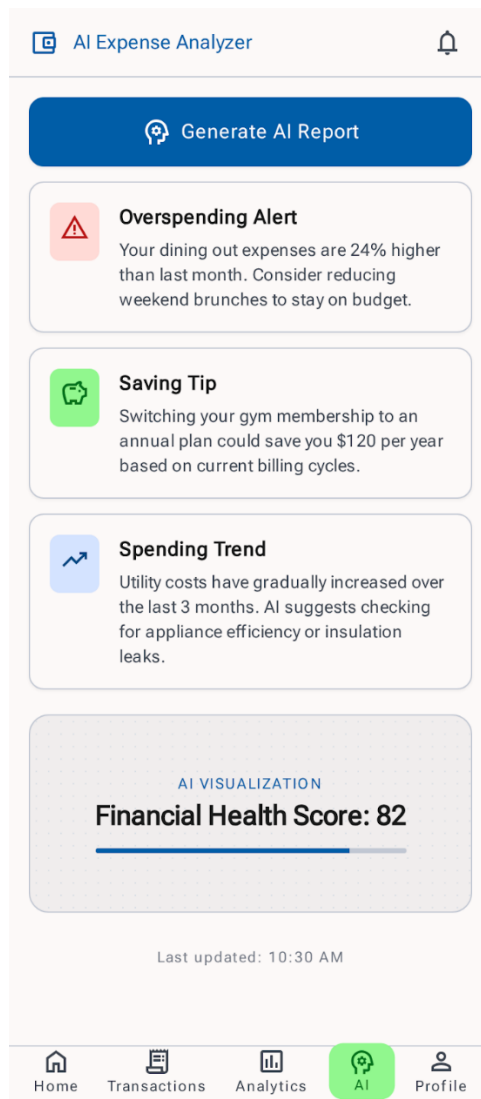
Analytics Screen:



Description:

A visualization hub that uses pie and bar charts to display spending patterns and category distributions.

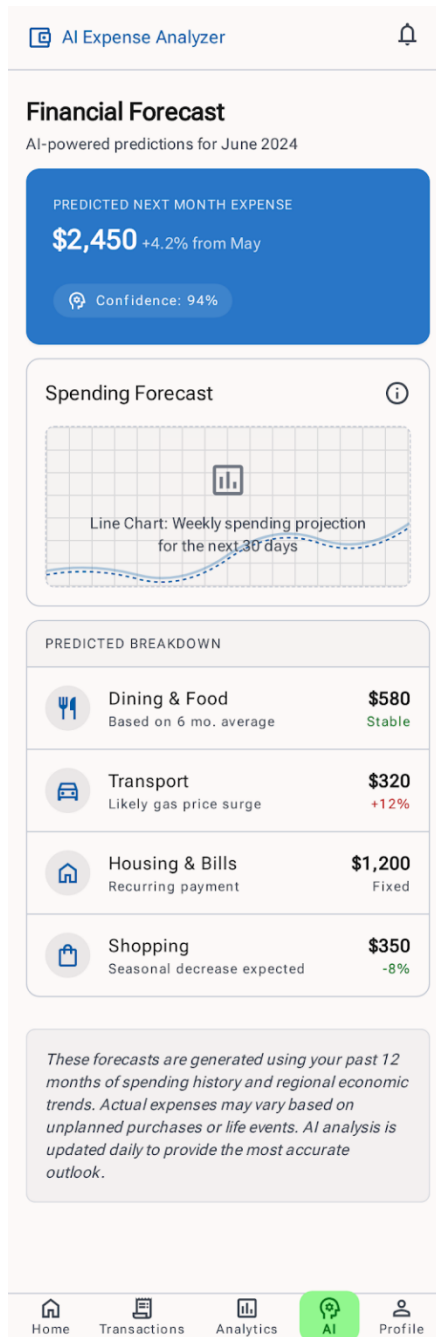
Ai_Insights Screen:



Description:

A dedicated section for actionable financial intelligence, including overspending alerts, saving tips, and the user's overall Financial Health Score.

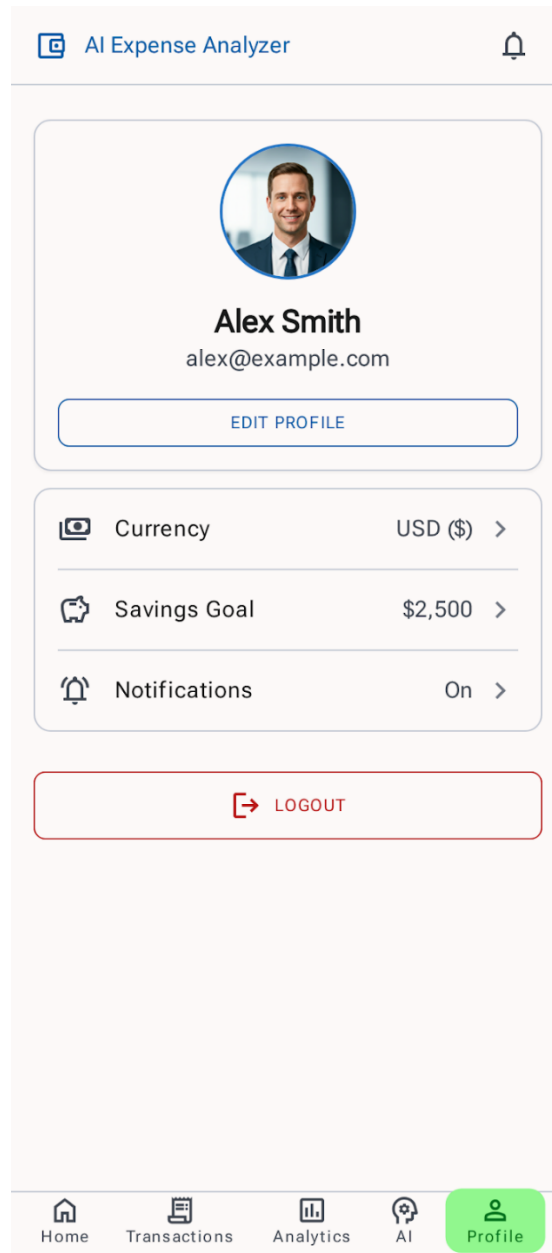
Spending_Forecast_Screen:



Description:

An analytical screen that leverages historical data to provide AI-powered budget predictions and expense breakdowns for the upcoming month.

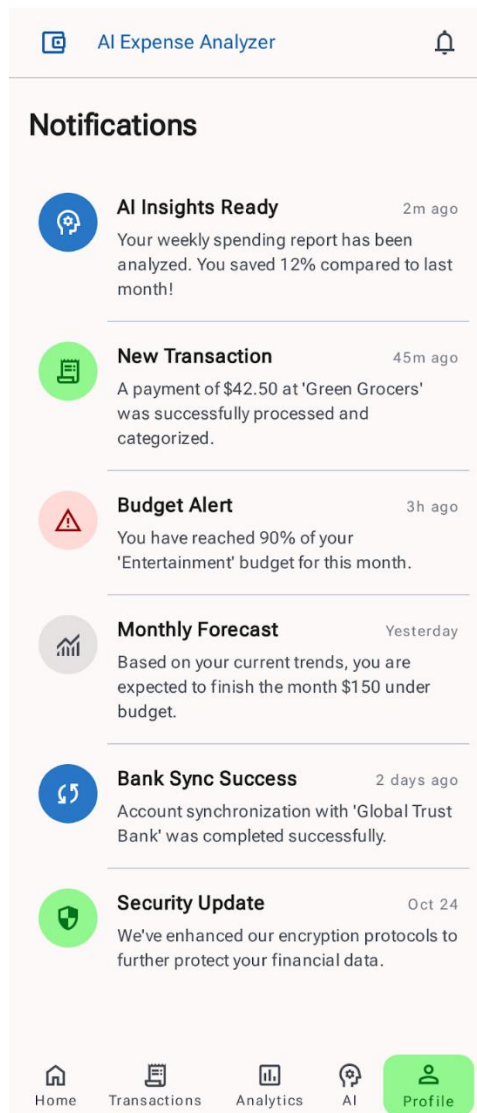
Profile Screen:



Description:

An interface allowing users to manage their account settings, update their personal profile, modify security credentials (like passwords), reset currencies, and adjust monthly savings benchmarks.

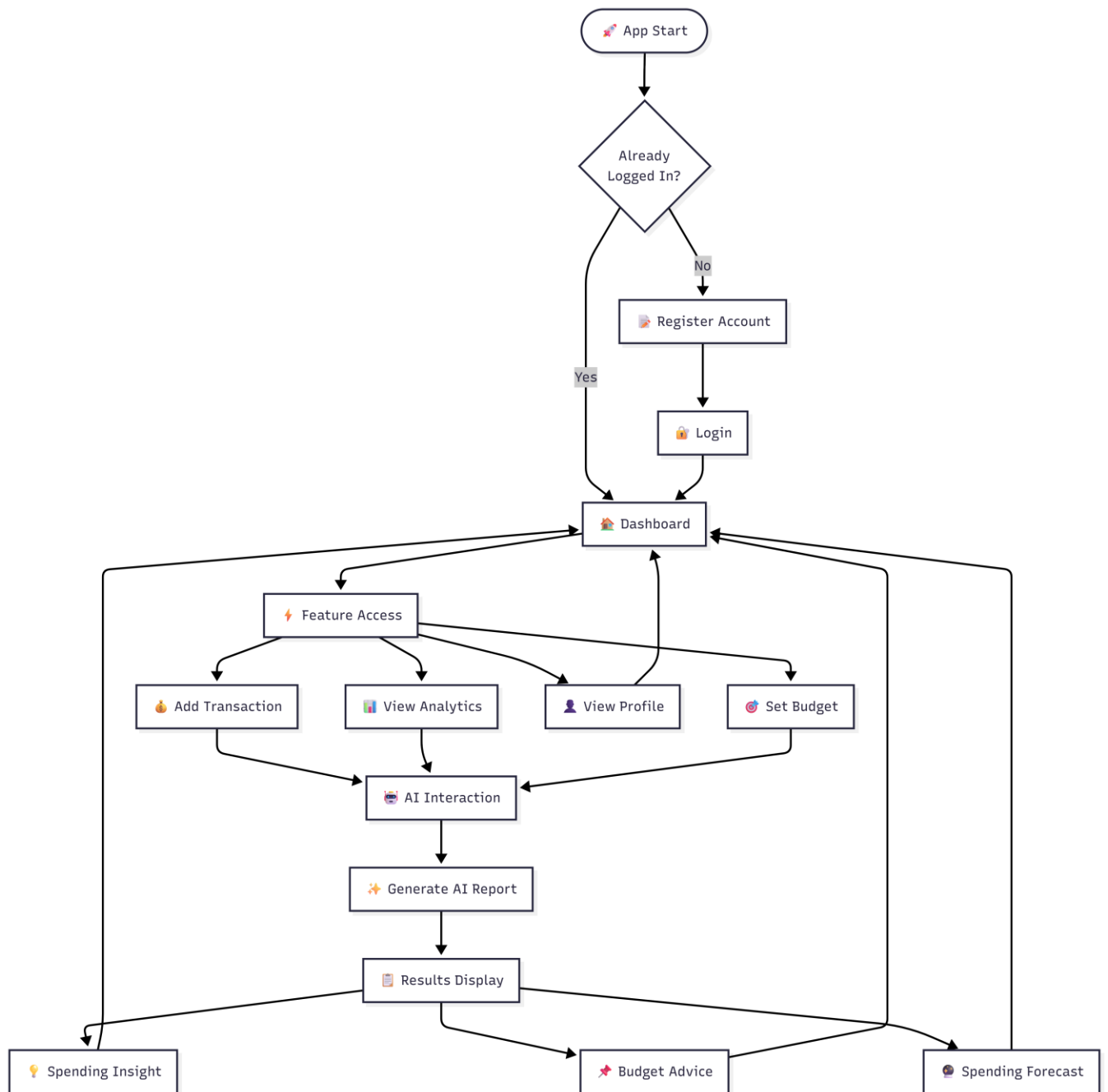
Notifications Screen:



Description:

A log of system alerts, informing users about budget threshold breaches, successful bank syncs, and newly generated AI reports.

User Flow Diagram:



Development Roadmap:

- **Week 06:** Backend Setup & Database Integration
- **Week 07:** Frontend UI Implementation
- **Week 08:** AI Model/Gemini API Integration
- **Week 09:** Feature Completion & Bug Fixing
- **Week 10:** Testing & Security Validation
- **Week 11:** Deployment

Team Task Distribution:

Member Name	Role/Responsibility
Naim Sheikh	Frontend Development
Ayon Banerjee	Backend Development
Md Shakinul Islam	Database Management
Afia Rahman	AI Integration & Documentation

Final Technology Stack:

- **Frontend:** Flutter
- **Backend:** Node.js, Express.js
- **Database:** MongoDB
- **AI Services:** Gemini API
- **Deployment:** Vercel/Render